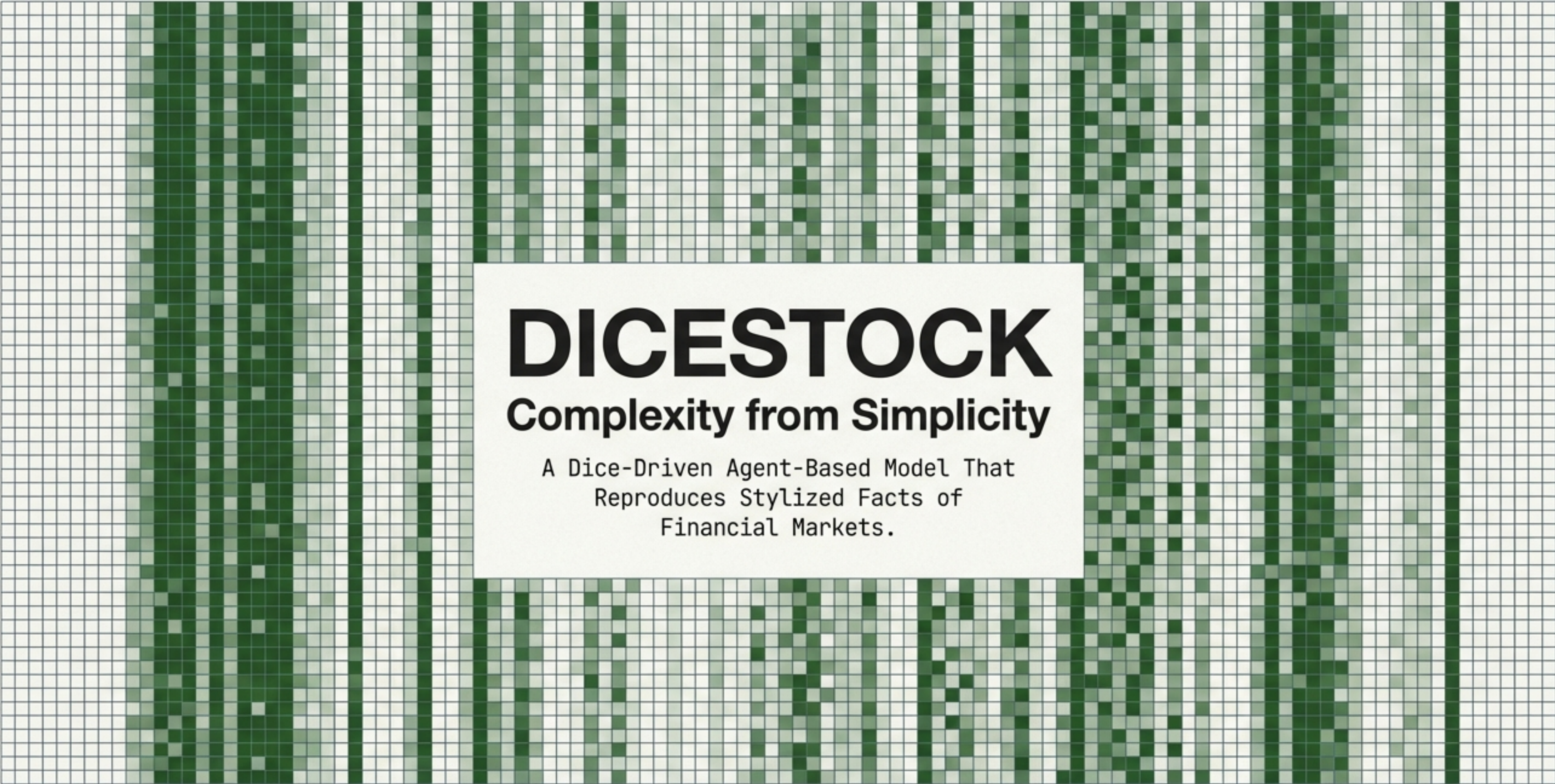


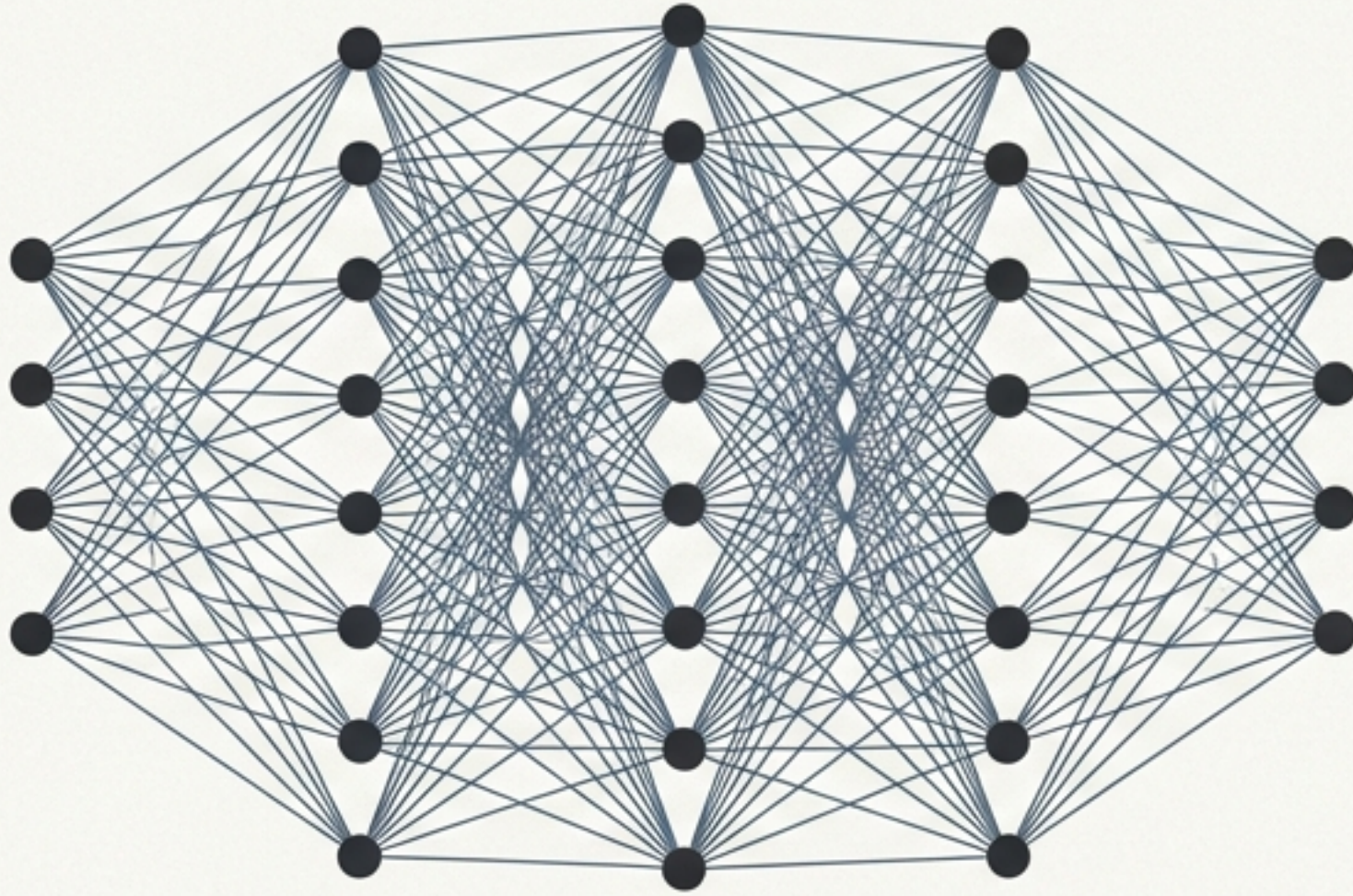


DICESTOCK

Complexity from Simplicity

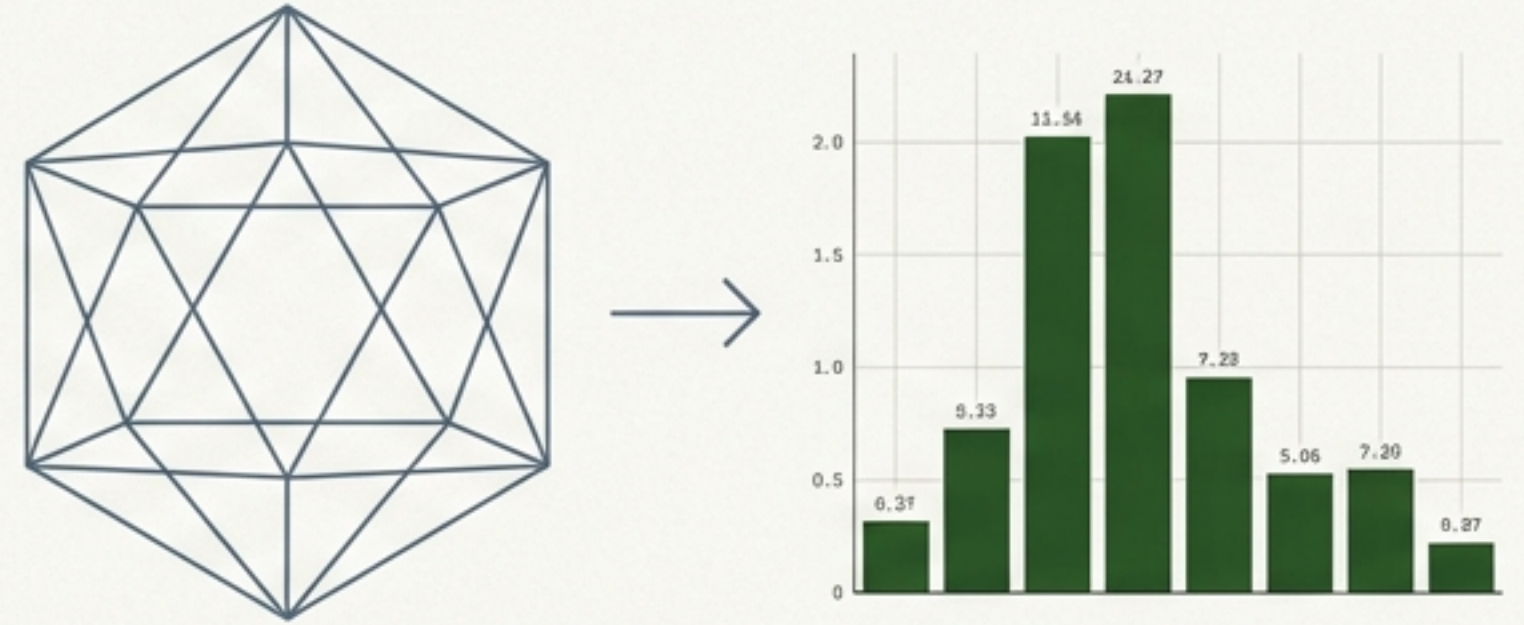
A Dice-Driven Agent-Based Model That
Reproduces Stylized Facts of
Financial Markets.

THE ORTHODOX APPROACH



High Agent Intelligence. Low Structural Transparency. Black Box Logic.

THE DICESTOCK APPROACH



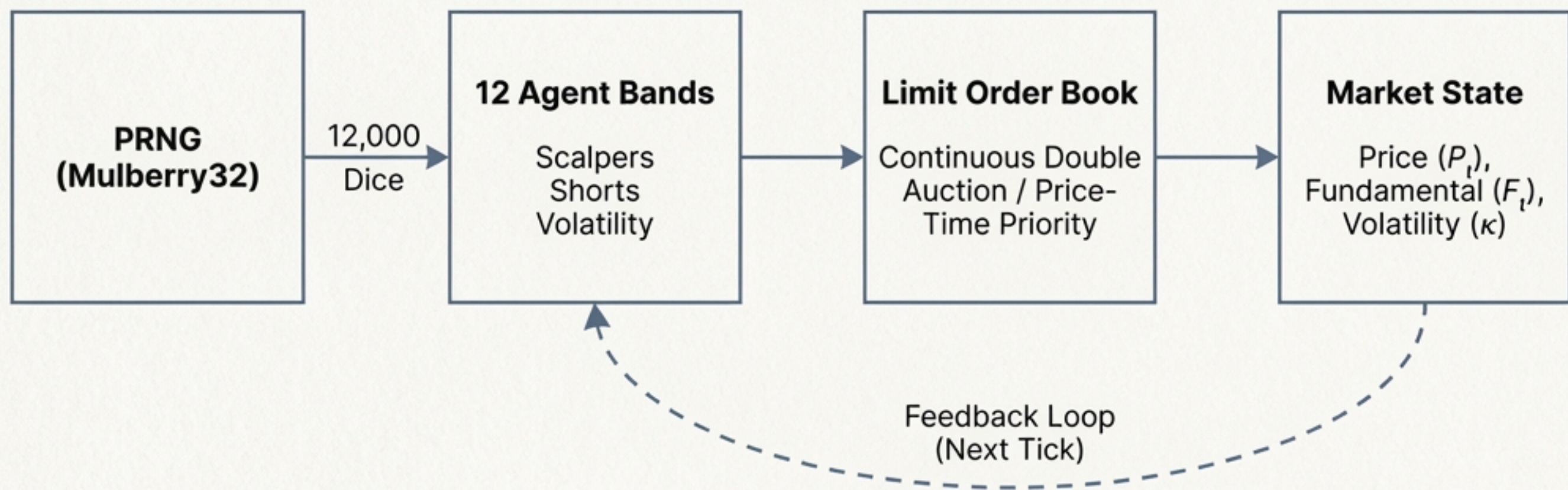
Zero Agent Intelligence. Maximal Structural Diversity. 12,000 Dice.

The Research Question:

Can realistic market dynamics—fat tails, volatility clustering, squeezes—emerge from agents driven *entirely* by random dice rolls, with no adaptive learning or pattern matching?

YES. DiceStock reproduces 9 of 10 stylized facts found in empirical finance.

DICESTOCK MODEL ARCHITECTURE



7.37 Million Ticks

75 Simulated Years

Reproducible on any IEEE 754 machine

THE AGENTS: DIVERSITY WITHOUT INTELLIGENCE

DIRECTIONAL

Bands 1-3: Multi-Timeframe

Logic: Scalp (1 tick), Swing (13 ticks), Position (65 ticks).

Action: Bullish > 14 / Bearish < 7.

STABILIZERS

Band 4: Fundamental Value Anchor

Band 6: Market Makers (Liquidity)

Band 12: Support/Resistance (Memory)

DISRUPTORS

Band 5: Short Sellers (Squeeze Logic)

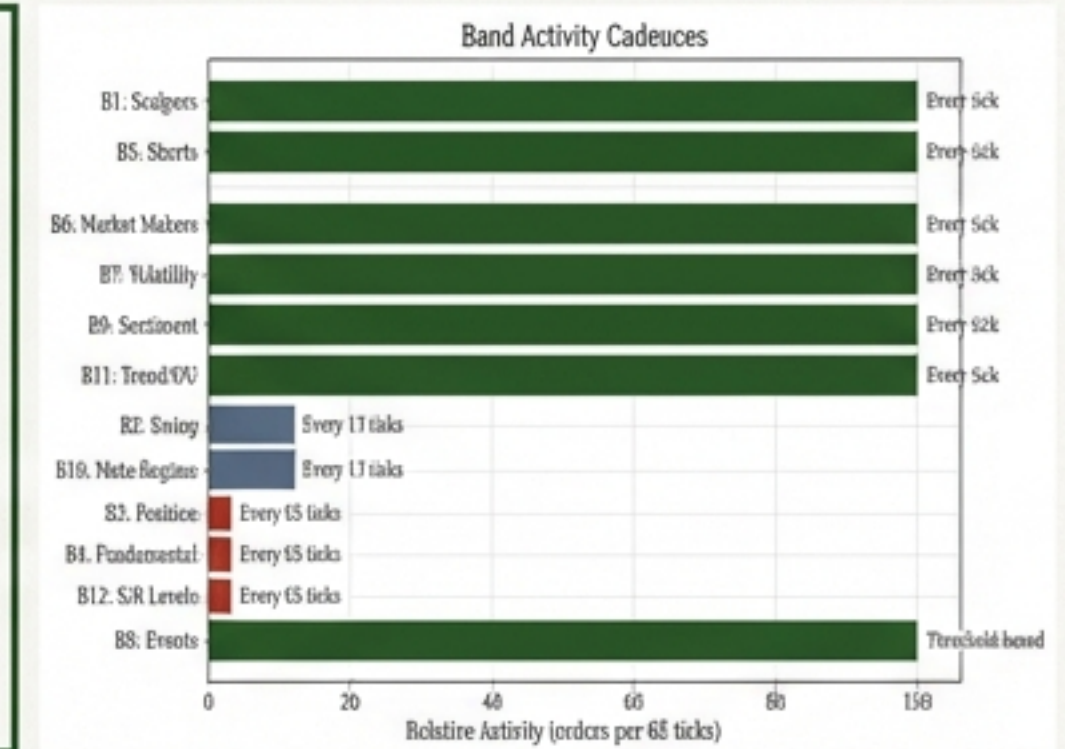
Band 8: Exogenous News Shocks

STATE DRIVERS

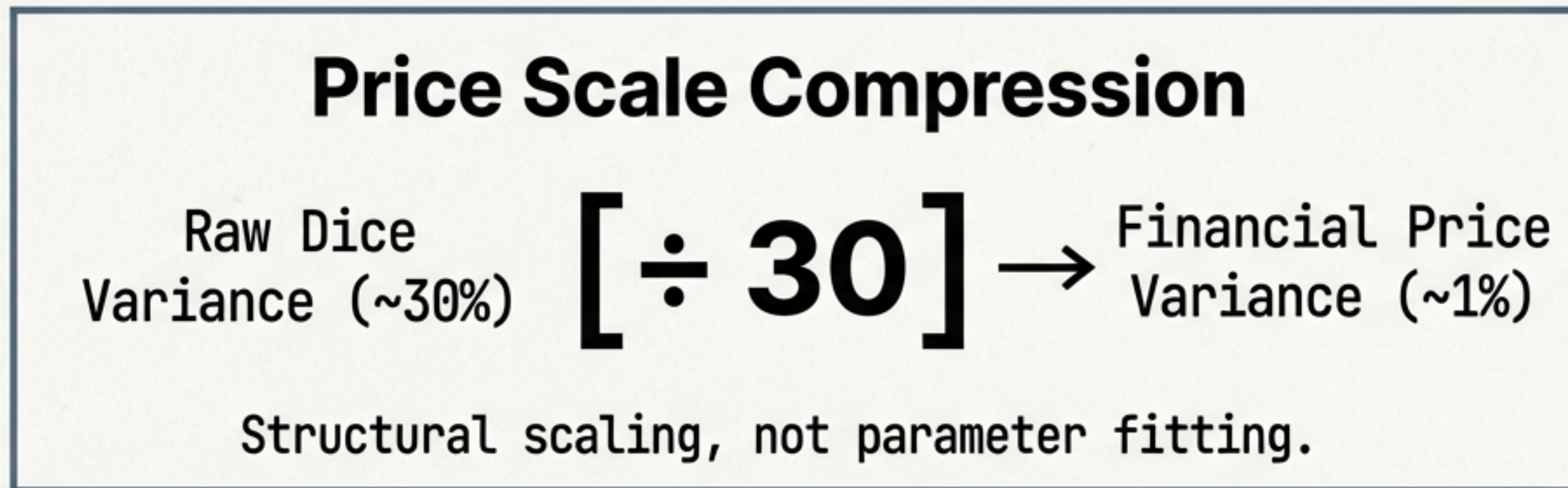
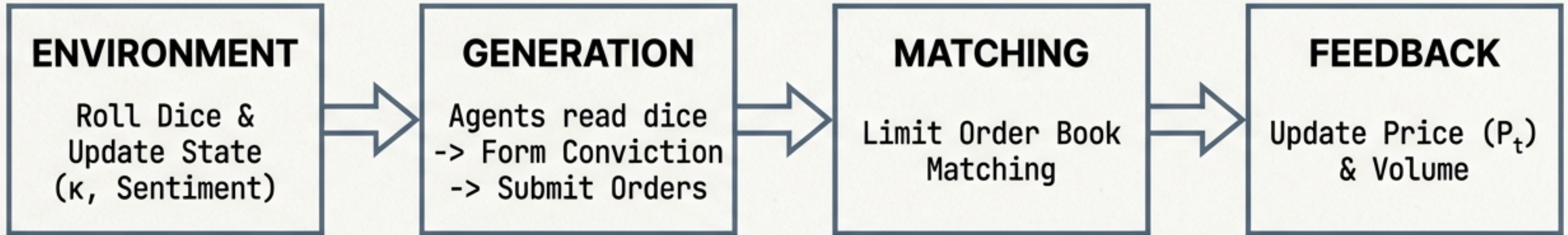
Band 7: Volatility Regime (κ)

Band 9: Crowd Sentiment

Band 11: Trend/OU Process



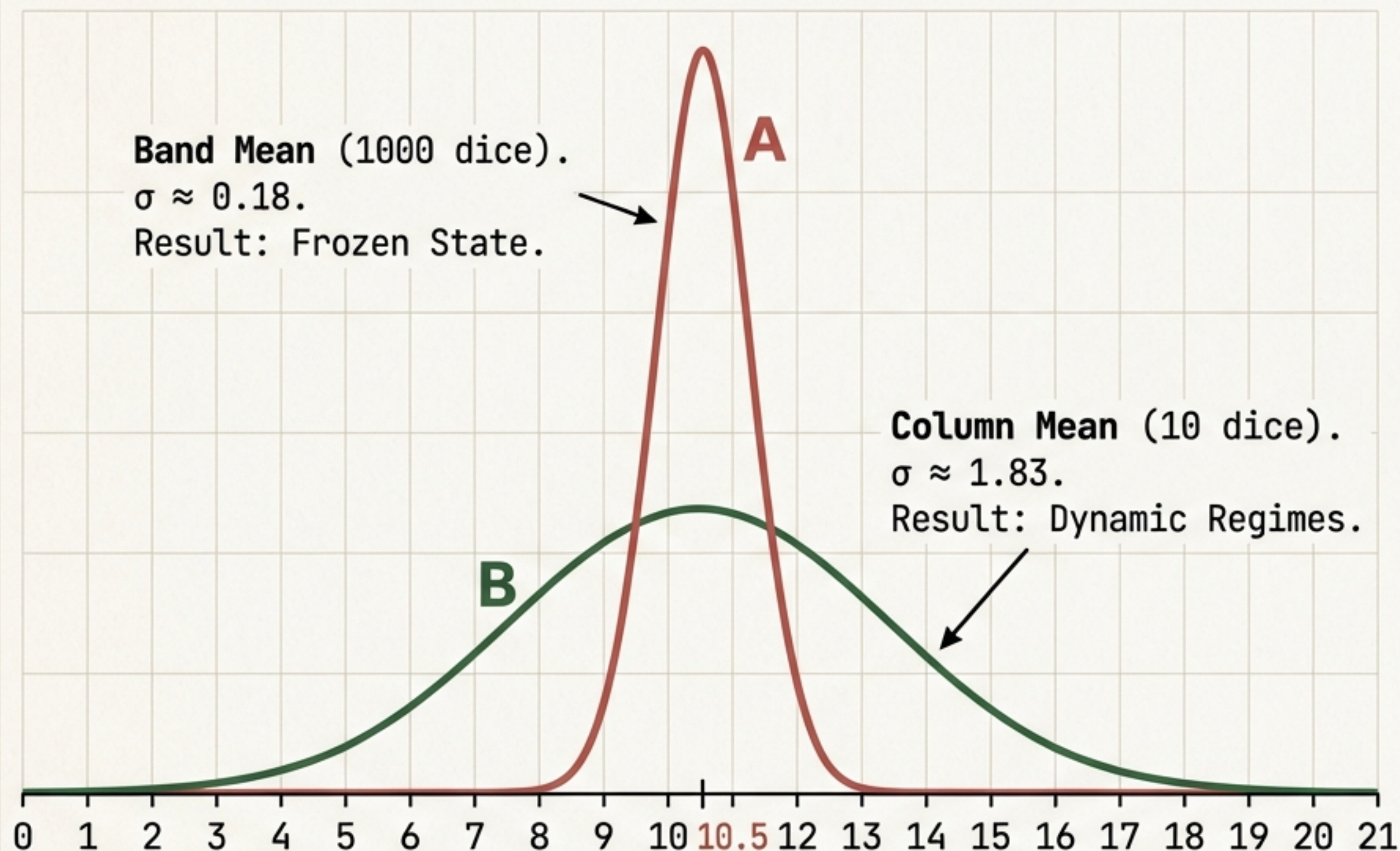
THE GEARBOX: FROM DICE TO PRICE



Microstructure: Orders expire after 100 ticks. Market Makers refresh every tick.

THE “AHA!” MOMENT: CALIBRATION & BUGS

Why variance matters.



v1 (Pre-Fix):

2/10 Stylized Facts.
Kappa stuck at 1.0.

v2 (Current):

9/10 Stylized Facts.
Rich volatility
regimes.

Takeaway: We fixed
statistical bugs in
signal processing,
we did not fit
parameters to data.

PRICE DYNAMICS: 75-YEAR TRAJECTORY



Start Price:
\$100.00

End Price:
~\$19,300

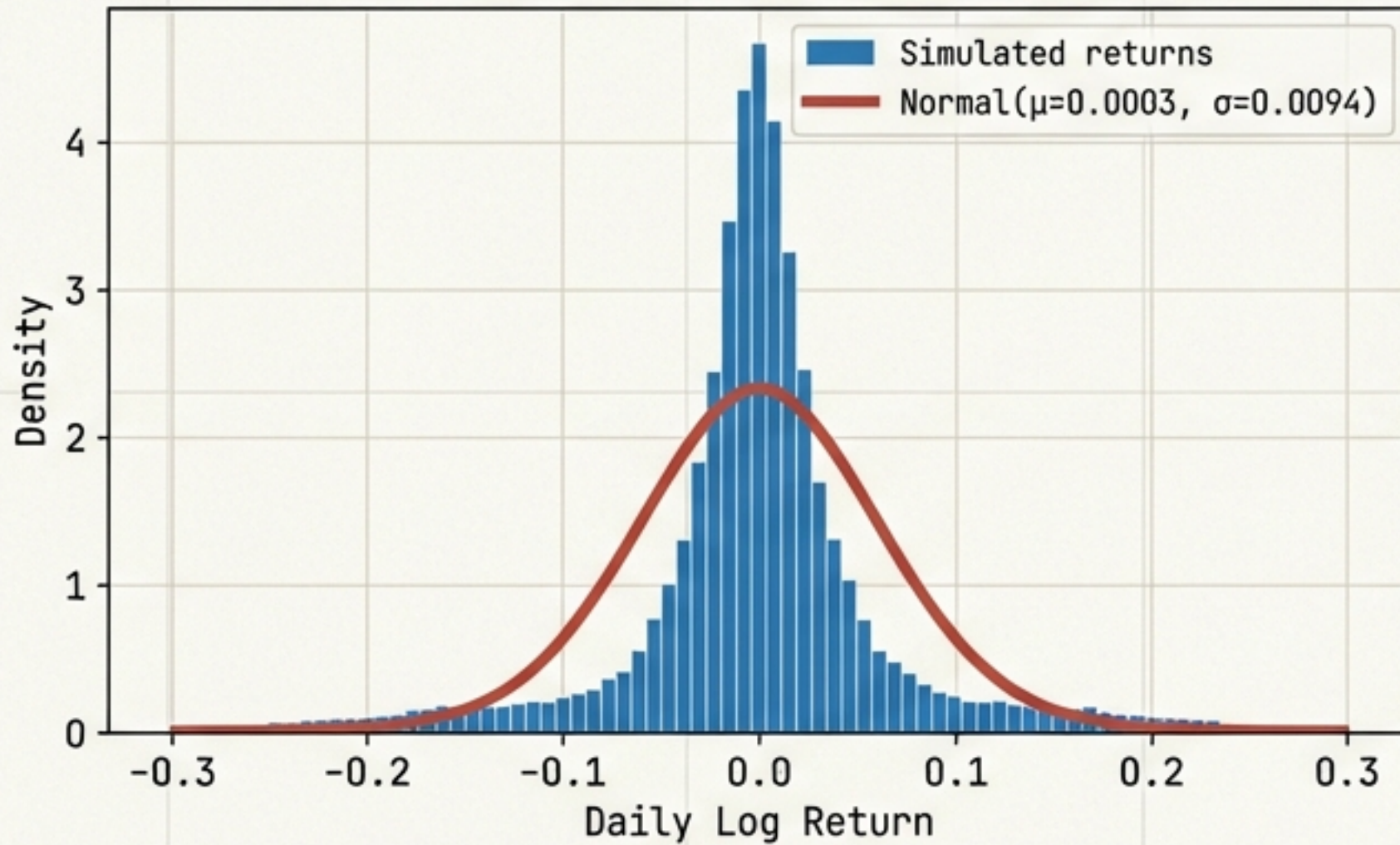
Ticks:
7,370,000

Note:
Price tracks
Fundamental
Value long-term,
but
deviates
significantly in
the short-term.

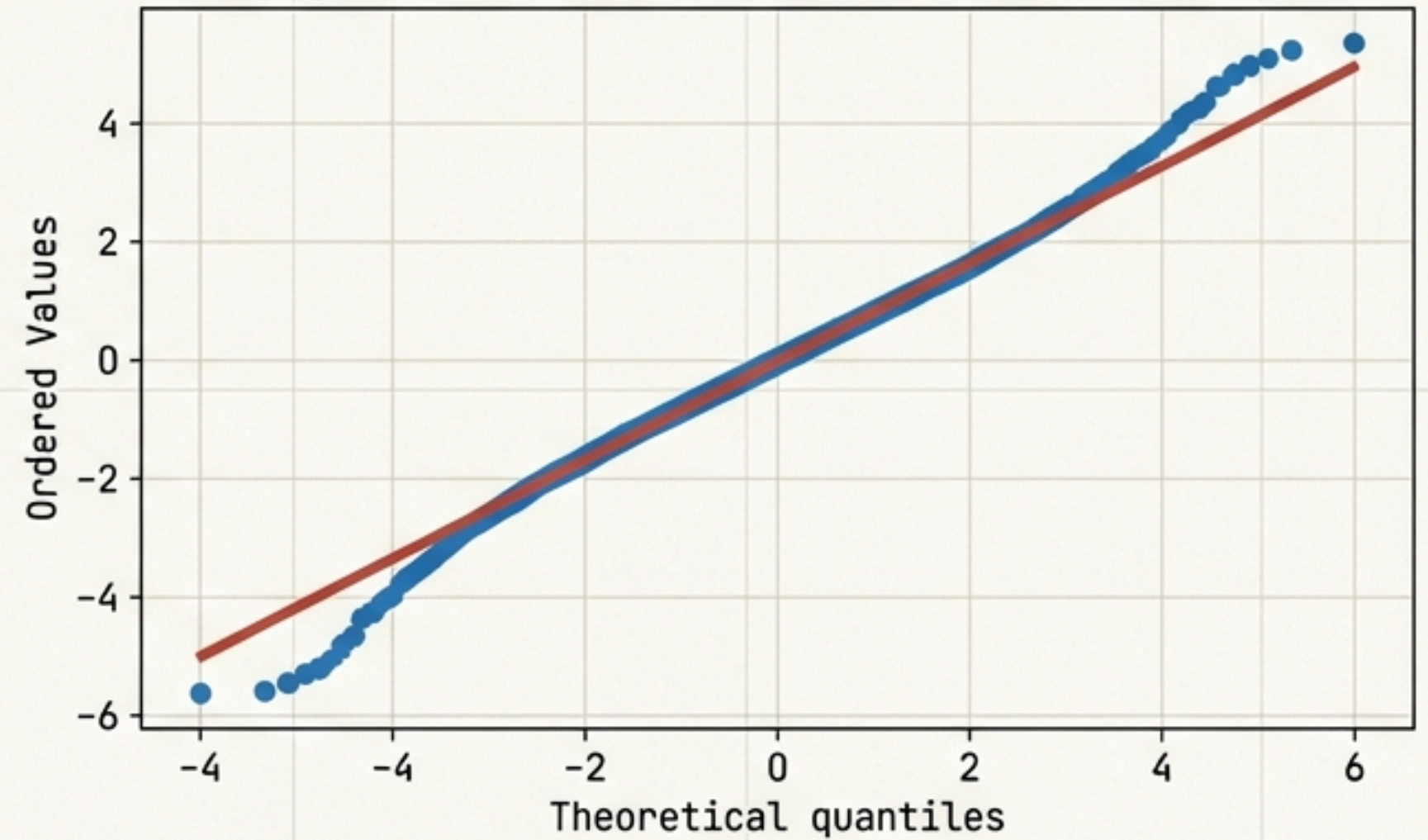
STYLIZED FACT #1: FAT TAILS

The market generates extreme events naturally.

Return Distribution vs Normal



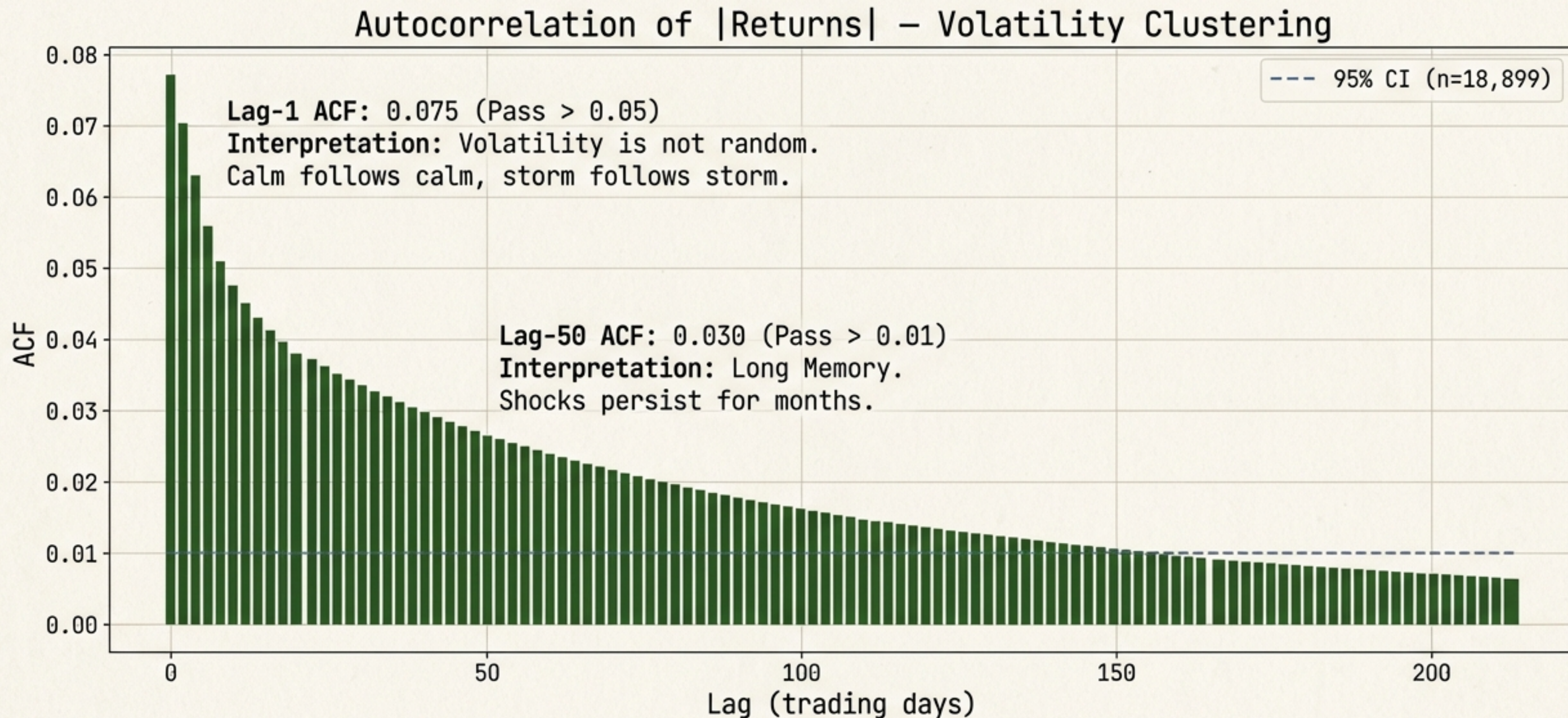
Q-Q Plot: Returns vs Normal Distribution



Excess Kurtosis: **13.91** (Normal=0) Hill Estimator: **2.45** Annualized Vol: **14.9%**

Drivers: Volatility Multiplier (κ) + Event Shocks + Squeeze Cascades.

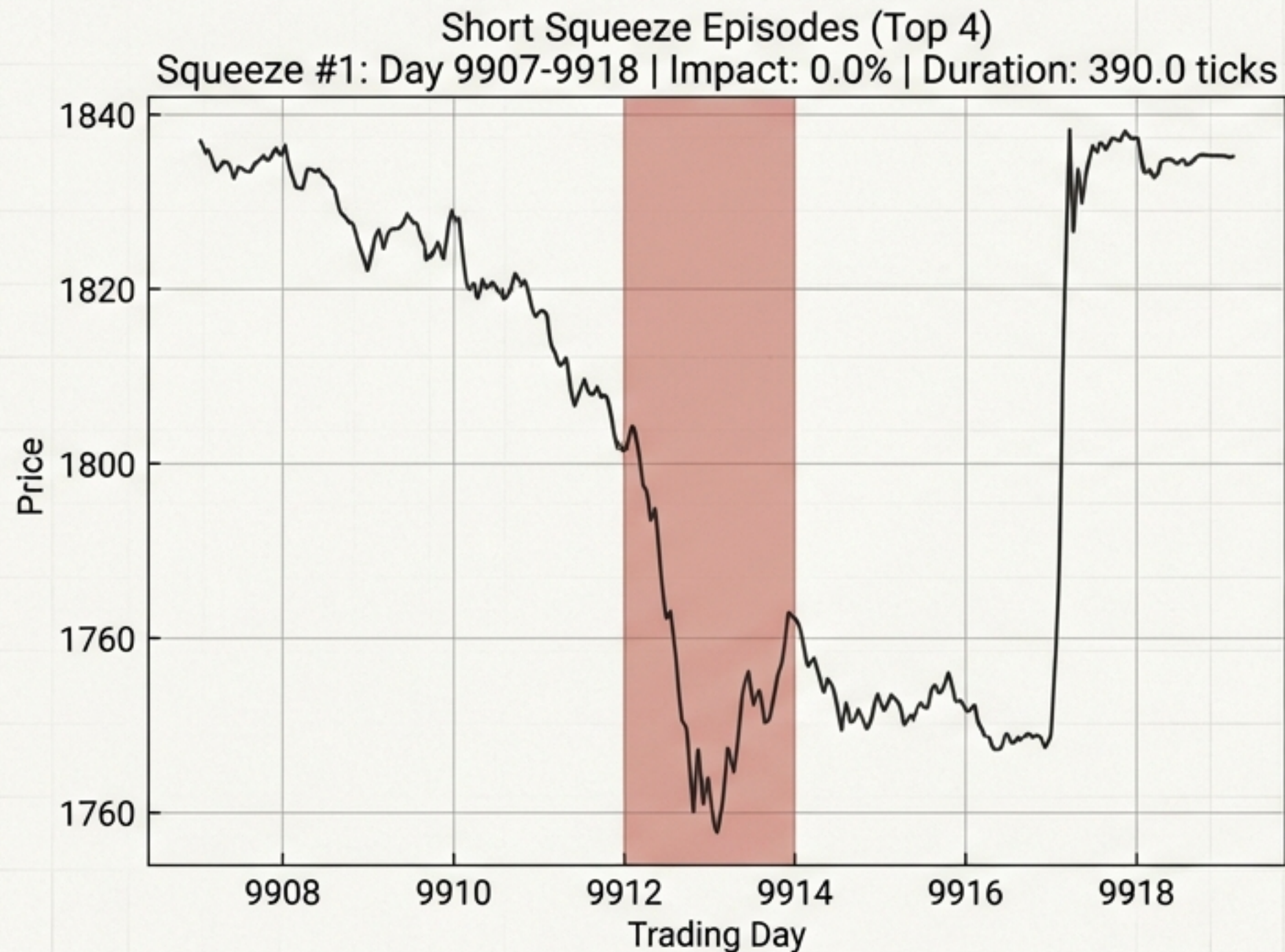
FACTS #2 & #3: CLUSTERING & MEMORY



Driven by 'Sticky Regimes' in Band 7 (κ) and Band 10 (Trend Persistence).

EMERGENT BEHAVIOR: THE SHORT SQUEEZE

Not scripted. Emerged from 4 conditions.

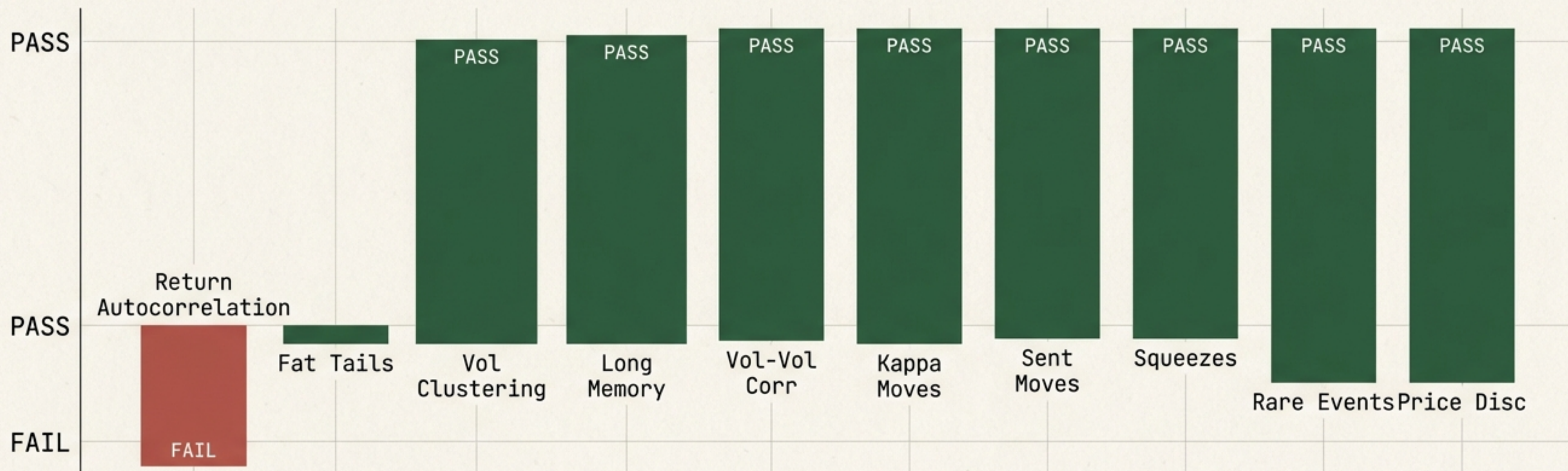


THE RECIPE FOR A SQUEEZE:

1. High Utilization: $> 45\%$ of float shorted.
2. Price Momentum: $> 3\%$ rise.
3. High Sentiment: > 0.57 (Crowd euphoria).
4. High Volatility: $\kappa > 1.2$.

Result: Forced covering caused a 390-tick vertical price spike.

THE SCORECARD: 9/10 STYLIZED FACTS

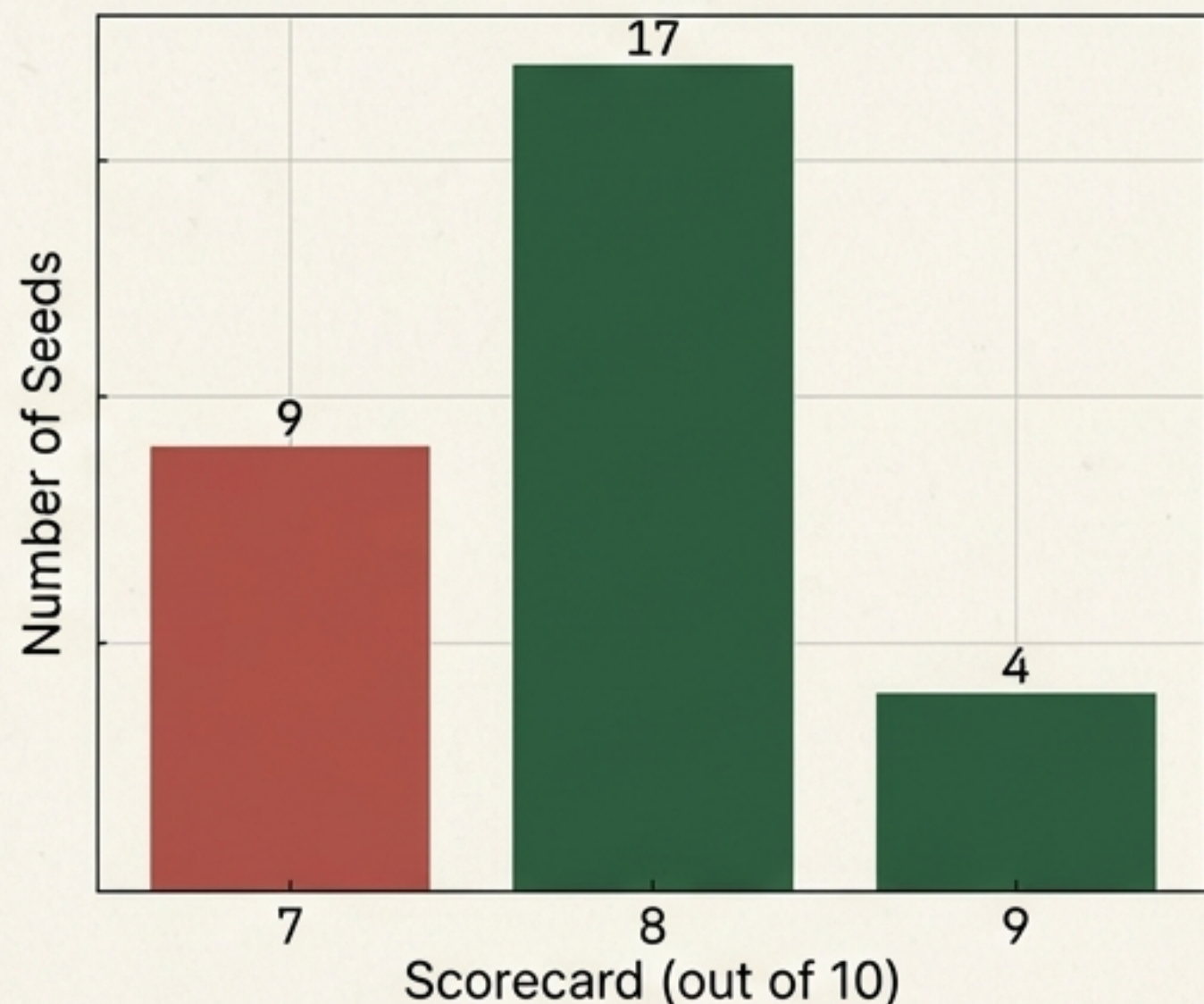


The Fail: Return Autocorrelation (-0.068 vs threshold ± 0.05).

Reason: Microstructure artifact. Aggressive Market Maker refresh causes mechanical bid-ask bounce.

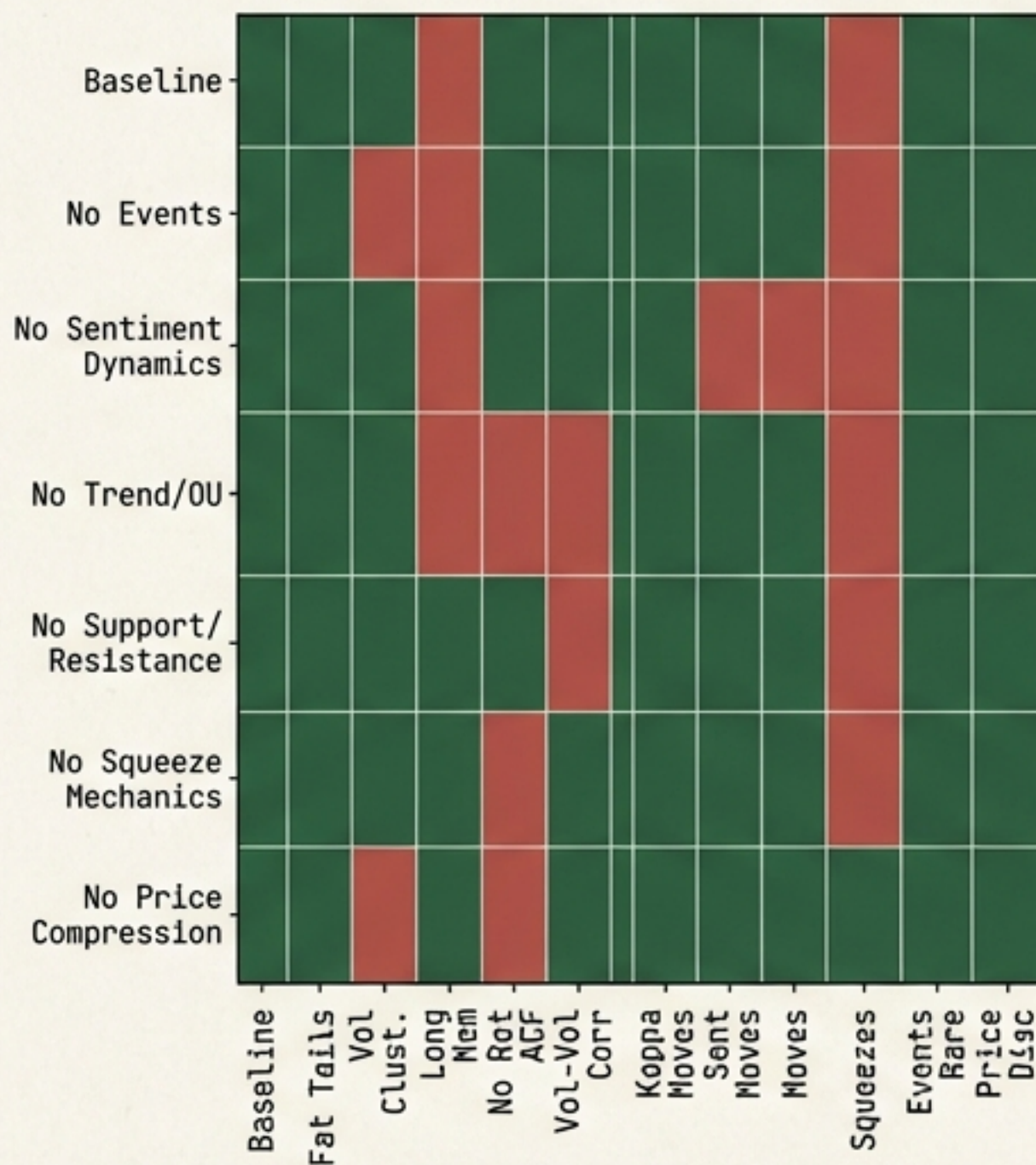
ROBUSTNESS & SENSITIVITY

Monte Carlo (30 Seeds)



100% of seeds pass Fat Tails.
Mean Score: 7.8/10.

Ablation Heatmap (What Matters?)



Load-Bearing:

Trend/OU (Band 11).
Removing this halves volatility.

Decorative:

Squeezes &
Support/Resistance.
Add realism but not
critical for
basic stats.

THE TURING TEST: DICESTOCK VS. S&P 500

Metric	DiceStock	S&P 500 (1950-2025)
Ann. Volatility	14.9%	15.8%
Hurst Exponent	0.540	0.541
Hill Estimator	2.45	2.91

Without seeing a real stock price, the Dice model converged on the exact volatility profile and fractal structure of the S&P 500.

TAXONOMY OF COMPLEXITY

	Cont-Bouchaud	Lux-Marchesi	Mike-Farmer	DiceStock
Agent Intel	Network	Switching	Zero	Zero (Dice)
Diversity	1 Type	2 Types	1 Type	12 Bands
Fitted Params	~5	~8	~6	0
Stylized Facts	3-4	5-6	4-5	9

We occupy a novel point in the design space: **Maximal Diversity, Minimal Intelligence.**

CONCLUSION

Stylized facts are a property
of *market structure*, not
agent sophistication.

- **Limitations:** Dead Zone in volatility regimes & mechanical autocorrelation.
- **Provocation:** If 12,000 dice can simulate the S&P 500, how much of real human trading is just noise filtered through a limit order book?